

ENGR. MUHAMMAD WAQAS

COMPREHENSIVE ENGINEERING, BIM & AI INNOVATION PORTFOLIO

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1. EXECUTIVE PROFILE & HYBRID ENGINEERING VISION

Engr. Muhammad Waqas represents a new paradigm in the Built Environment: a Civil Engineer, Manager Estate (Technical) at REDAMCO (Ministry of Railways) and Senior Infrastructure Specialist with 15+ years of hands-on field, infrastructure, and project management leadership who has systematically integrated Building Information Modeling (BIM / VDC), ISO 19650 Information Management, and Applied Artificial Intelligence (LLM & Agentic Workflows). With a proven background bridging corporate railway asset monetization, client-side megaprojects (Pakistan Railways), GCC industrial mega-structures (Sohar Steel Mill Oman), and multi-story residential/commercial developments, Waqas designs, executes, and automates modern construction workflows.

2. FEATURED AI & DIGITAL AEC INNOVATION PROJECTS

PROJECT 01: AI-Architect — Generative AI Layout & BIM Engine

Domain: AI Engineering, Generative Architecture & CAD Automation | GitHub Repository: <https://github.com/engrwaqas034-AI/AI-Architect>

- **Concept & Innovation:** Engineered a cutting-edge generative AI platform that automates 2D/3D architectural floorplan generation based on plot dimensions, setback rules, and functional zoning requirements.
- **Technical Stack:** Python, Custom CSP (Constraint Satisfaction Problem) Solver, SVG Parametric Engine, RAG Knowledge Base, HTML5/CSS3 Interactive UI.
- **Key Functionalities:** Calculates zoning envelopes, auto-places living/dining/bedroom spaces with zero overlapping, verifies setback compliance, and exports standard DXF/BIM coordinate structures.
- **Impact:** Reduces preliminary architectural floorplan drafting time from days to seconds while eliminating spatial design errors.

PROJECT 02: Construction_Claim_Analyzer — LLM & RAG Delay Claim Intelligence

Domain: AI Engineering, Claims & Dispute Resolution, Construction Law | GitHub Repository: https://github.com/engrwaqas034-AI/Construction_Claim_Analyzer

- **Concept & Innovation:** Developed an intelligent LLM & RAG (Retrieval-Augmented Generation) analytical system specialized in evaluating construction delay claims, Extension of Time (EOT) applications, and contractual disputes.
- **Contractual Knowledge Base:** Embedded FIDIC standard contracts, project baseline schedules (Primavera P6), daily site reports, and variation orders into a searchable vector database.
- **Automated Analysis:** Performs instant delay attribution, critical path disruption checks, and contract clause compliance verification, outputting structured legal/engineering risk scores.
- **Impact:** Empowers claims consultants and project managers to audit complex contractual claims in minutes rather than weeks.

PROJECT 03: AI BOQ & Quantity Surveying Generator

Domain: AI Engineering, Quantity Surveying & Cost Estimation | Repository: Integrated AEC AI Tool Pipeline

- **Concept & Innovation:** Built an automated AI-driven pipeline for extracting Bill of Quantities (BOQ), material demands, and structural steel/concrete quantities directly from architectural inputs.
- **Technical Features:** Parses structural parameters, calculates concrete volumes, bar bending schedules (BBS), and formwork areas, linking directly with cost databases.
- **Impact:** Eliminates manual take-off errors, drastically accelerating pre-construction estimation and tender preparation.

3. FEATURED BIM & VDC (BUILDING INFORMATION MODELING) PROJECTS

PROJECT 04: Construction of Government Primary School KPK (Complete BIM Delivery)

Location: Khyber Pakhtunkhwa, Pakistan | Role: Lead BIM Engineer | Tools: Revit, Navisworks Manage, Primavera P6, Twinmotion

- Integrated 3D BIM Modeling: Created high-detail architectural and structural models in Autodesk Revit, ensuring full multi-disciplinary clash detection.
- 4D Construction Simulation: Linked 3D Revit models with Primavera P6 schedules in Navisworks Manage for step-by-step visual construction sequencing.
- 5D Cost & Quantity Tracking: Generated automated BOQs, material schedules, and progress cost tracking directly from the BIM database.
- Visualization & Impact: Rendered photorealistic 3D walkthroughs in Twinmotion, achieving a +15% increase in site construction efficiency.

PROJECT 05: Modern Residential Villa (3D BIM & Real-Time Visualization)

Location: Rawalpindi / Islamabad | Role: BIM Designer & Coordinator | Tools: Revit, Twinmotion, AutoCAD

- 3D Spatial Design: Developed complete 3D BIM models emphasizing space optimization, natural illumination, and modern architectural aesthetics.
- Visualization & VR Walkthroughs: Rendered real-time client presentations in Twinmotion featuring realistic lighting, material textures, and spatial flows.
- Construction Documentation: Extracted precise 2D architectural plans, elevations, sections, and structural reinforcement details.

PROJECT 06: Civil 3D Road Infrastructure & Earthwork Modeling

Location: Infrastructure Sector | Role: Civil Infrastructure Specialist | Tools: Autodesk Civil 3D

- Surface & Topography: Generated accurate 3D terrain surfaces from existing survey point clouds and GIS datasets.
- Corridor Design: Designed road alignments, vertical profiles, assemblies (lanes, shoulders, ditches), and 3D corridors.
- Earthwork Optimization: Calculated precise cut-and-fill volumes, optimizing grading design to minimize earthwork costs.

4. MAJOR CIVIL & INFRASTRUCTURE PROJECTS (15+ YEARS FIELD EXPERIENCE)

REDAMCO (Railway Estate Development & Marketing Company) — Ministry of Railways (Dec 2025 – Present)

Role: Manager Estate (Technical) | Public Sector Corporate & Asset Management

- Performs technical appraisals, execution feasibility reviews, and engineering evaluations for Ministry of Railways real estate developments.
- Assists Senior Manager Estate and Business Development in land monetization strategies, commercial joint ventures, and revenue optimization.
- Conducts structural/infrastructural risk assessments and lease compliance audits on commercial railway assets across Pakistan.

Pakistan Railways Infrastructure & Railway Network Projects (May 2018 – 09th Dec 2025)

Role: Inspector of Works / Infrastructure Engineer | Client: Pakistan Railways (Public Sector)

- Oversees maintenance, track renewal, station rehabilitation, and bridge safety across major railway corridors.
- Utilizes GIS land surveying to clear right-of-way (ROW) encroachments and map railway land boundaries.
- Prepares structural designs, BOQs, technical specifications, and liaises with government agencies for project sanctions.

Housing Directorate GHQ — Askari 5, 11 & 14 Projects (Oct 2013 – Jul 2016)

Role: Site Engineer (Civil) | Contractor: Construction Invadar Pvt Ltd

- Supervised construction of 2xG+4 residential flats, luxury villas, internal road networks, drainage, and utilities.
- Managed IPCs, Running Account Bills (RARs), deviation orders, Bar Bending Schedules (BBS), and subcontractor payments.

Sohar Steel Mill Structure & Residential Villas, OMAN (May 2011 – Sep 2013)

Role: Site Engineer (Civil) | Company: Al Raak International LLC (Muscat & Sohar, Oman)

- Executed heavy structural steel mill framing, precast concrete erection, slope protection, and access ramps in Sohar, Oman.
- Supervised batching plant concrete deliveries, site grading, paving, and measurement verification.

DHA Emaar Housing Society Infrastructure Project (Mar 2017 – Jan 2018)

Role: Site Engineer (Civil) | Contractor: Shah Zaman Constructions Pvt Ltd

- Managed site execution, baseline scheduling in Primavera P6, material procurement, BBS, and interim payment claims.